

Pakistan Poverty Lens: Satellite-Driven Indicator Mapping for Smart Development Policy

D. PROJECT EXECUTIVE SUMMARY . Describe the proposed research using (not more than 3000) characters

In Pakistan, conventional poverty surveys often lag behind rapidly changing realities, especially in remote or underserved areas. This project brings a fresh perspective: using satellite-based observations to reveal economic hardship in fine detail—across both urban neighborhoods and rural villages.

The platform will interpret indicators like housing patterns, infrastructure access, land use, vegetation, and rooftop solar presence. By estimating solar panel distribution via imagery, the system can also inform energy planning—highlighting where self-generated electricity might offset hardship or supply gaps.

To make this powerful and flexible, each extracted feature—building density, vegetation cover, solar panels, roads—is handled by its own microservice. These building blocks are organized and cataloged in a Microservices Reusability Index (MSRI), enabling rapid reuse and adaptation as new analytic needs arise.

The system isn't just a tool for analysts—it's a national asset. We will deliver these microservices to the Pakistan Bureau of Statistics (PBS) and integrate all outputs with the newly inaugurated National Big Data Portal, launched by Minister Ahsan Iqbal in mid-June 2025 as Pakistan's centralized hub for public data.

Through this connection, policymakers, researchers, and planners will access poverty maps, solar estimates, and infrastructure indicators all updated in near real-time.

The system operates in cycles—ingesting new imagery, extracting features, generating maps, learning from discrepancies, and refining outputs—all under strong governance and with modular, reusable components.

In essence, this project builds a resilient, transparent, and AI-driven platform for poverty detection and energy resource mapping. It's modular, evolving, and integrated into Pakistan's national data infrastructure. Most importantly, it equips policymakers and development partners with clear, trustworthy insights to drive equitable, evidence-based action across the country.

E1. PROJECT OBJECTIVES

This initiative aims to build a **trusted AI-powered platform** that reveals hidden poverty and energy potential across Pakistan by combining satellite imagery, advanced analytics, and governance support. It will:

- **Illuminate poverty and energy landscapes by estimating rooftop solar panel distribution alongside socio-economic indicators such as building density, infrastructure access, and land cover—enhancing both poverty detection and energy planning.**

- **Develop a predictive poverty estimation model** integrating satellite-derived indicators with AI analytics.
- **Develop a microservices-based framework that enables reusable feature extraction services (e.g., solar detection, building segmentation, vegetation mapping) evaluated using reusability indices.**

By achieving these objectives, the project lays a foundation for a data-rich, transparent, and modular ecosystem—advancing poverty alleviation and energy access in Pakistan, while setting a standard for resilient, AI-powered public infrastructure.

E2. PROJECT DESCRIPTION

Pakistan confronts two critical and interlinked challenges: accurately identifying pockets of poverty across its varied urban and rural landscape, and understanding the decentralized adoption of rooftop solar installations that affect both energy demand and supply. Traditional methods—such as household surveys—are costly, slow, often outdated, and miss dynamics in rapidly changing or remote areas. Meanwhile, burgeoning rooftop solar growth, while promising energy access, remains largely undocumented at scale, complicating planning for supply balancing and grid resilience.

To address this, we propose a unified, scalable solution that leverages satellite imagery to detect signs of economic deprivation and to estimate the spread of solar panels at a granular level. The core challenge is to design a system of modular microservices—each dedicated to extracting a distinct feature from imagery, such as building density, vegetation indicators, infrastructure gaps, and solar panel presence—while ensuring these microservices are catalogued and shareable through a proprietary Microservices Reusability Index (MSRI).

Another essential dimension is integration and governance. Outputs must feed directly into the Pakistan Bureau of Statistics (PBS) and the recently inaugurated National Big Data Portal, offering policymakers timely, layered insights to inform social interventions and energy planning.

In summary, the project must resolve this multifaceted problem:

How can satellite imagery be transformed into actionable insights that map poverty and solar uptake at fine resolution across diverse contexts?

How can feature extraction be modular, reusable, and maintainable, via a governed microservices architecture (MSRI)?

How can these insights be embedded seamlessly into national decision-making channels like PBS and the Big Data Portal?

Solving this challenge promises to empower policymakers, educators, NGOs, and energy planners with evidence-based tools that illuminate where help is needed most—and where renewable energy might already be reshaping communities.

E3. PROBLEM STATEMENT

Conventional poverty assessment methods in Pakistan rely heavily on household surveys that are infrequent, costly, and slow to capture the rapidly changing realities of economic hardship, particularly in remote and underserved regions. This time lag results in outdated insights that hinder effective policymaking and resource allocation. Moreover, existing approaches lack the ability to integrate diverse indicators such as housing quality, infrastructure access, vegetation, and renewable energy adoption, which are increasingly vital for understanding poverty and planning sustainable development.

There is, therefore, a critical need for a resilient, transparent, and AI-driven system that can harness satellite imagery and big data technologies to deliver near real-time, fine-grained poverty and energy resource maps. Such a platform must not only detect socioeconomic disparities but also be modular, reusable, and seamlessly integrated with Pakistan's national data infrastructure to empower policymakers, researchers, and development partners in making evidence-based, equitable decisions.

E4. METHODOLOGY

To transform satellite imagery into precise, actionable insights on poverty and solar energy distribution, we follow a structured, iterative workflow aligned with the principles of CRISP-DM and the Data Science Lifecycle (DSLCL). The process comprises six integrated stages:

1. Data Acquisition & Understanding

Satellite imagery scraping: We collect up-to-date optical and multispectral imagery from providers such as Google Earth, Sentinel-2, or PlanetScope, ensuring broad spatial coverage across urban and rural areas.

Exploratory analysis: We inventory image quality, seasonal variation, and relevant metadata (e.g., acquisition date, band composition), and align these with ground-truth labels or survey data where available.

2. Data Preparation & Labeling

Ground-truth generation: Using a combination of expert annotation, crowd-sourced labeling, and existing statistical overlays, we label structures (buildings, roads), vegetation, water bodies, and rooftop solar panels.

Preprocessing: Image data is normalized, resampled, and tiled for efficient processing. A version-controlled labeling workflow ensures consistency among labelers and traceability of decisions.

3. Modeling & Training

Model selection: We employ state-of-the-art segmentation architectures such as U-Net, DeepLabV3, FPN, and PSPNet to detect features like building density, vegetation cover, roads, and solar panels.

Iterative retraining: Models are trained and fine-tuned using transfer learning and domain adaptation methods, with continuous validation against hold-out labeled datasets to guard against overfitting.

4. Evaluation

Performance metrics: Each model's accuracy is assessed using Intersection over Union (IoU) and Dice Coefficient scores to evaluate the quality of segmentation masks.

Cross-validation: We test generalizability across different regions—urban vs. rural, coastal vs. inland—to ensure robustness.

5. Infrastructure & Scaling

High-performance computing: Model training and inference tasks are run on HPC infrastructure, leveraging NVIDIA RTX 4090 GPUs for rapid, large-scale processing of imagery tiles.

Microservices architecture: Each feature extraction (e.g., solar detection, building mask) is packaged as an independent microservice. These services are cataloged via our Microservices Reusability Index (MSRI), facilitating reuse and orchestration in complex workflows.

6. Deployment & Governance

Integration & deployment: Validated microservices and feature outputs are integrated into national systems—most notably, the Pakistan Bureau of Statistics (PBS) and the National Big Data Portal—enabling real-time access for policymakers and analysts.

Governance framework: Embedded stakeholders like data owners and data stewards oversee quality, maintain data lineage, and manage access controls, ensuring transparency, traceability, and compliance.

F. PROJECT OUTCOMES & BENEFITS

1. Solar Energy Insights for Decentralized Planning

High-resolution imagery and AI methods can accurately detect rooftop solar installations—even in complex, large-scale settings—with up to 97.8% precision.

These solar detection results feed into energy planning by estimating distributed generation capacity and identifying unmet energy needs.

2. Operational Efficiency & Cost Savings

Satellite-based solar monitoring dramatically reduces costs and logistical burdens compared to traditional surveys or on-site inspections. AI-powered detection enables real-time monitoring, early issue detection, and efficient maintenance, enhancing the lifespan and performance of solar installations.

3. Accelerated Aid & Development Response

Analogous applications—such as disaster response—have leveraged satellite AI systems to deliver targeted aid faster than traditional methods, improving outcomes and preserving dignity for vulnerable communities. Our solution holds similar potential in both poverty mitigation and energy access planning.

4. Scalable, Modular Intelligence Through Microservices

Each satellite-derived feature—for example, building density, rooftop solar, or vegetation—is served through independent, reusable microservices cataloged via the Microservices Reusability Index (MSRI). This design accelerates future developments, encourages rapid adaptation, and promotes collaboration across agencies.

5. Sustainable Foundations for Future Innovations

This platform lays groundwork for adding new feature modules (e.g., disease spread, school access, water resources) in future iterations, amplifying both its social impact and technological relevance over time.

Summary: The project yields multidimensional benefits—equipping policymakers with detailed, trustworthy poverty and energy insights; enabling smarter energy strategies and solar deployment; driving efficient, data-led development; and embedding governance, scalability, and fairness throughout its lifecycle. Supported by empirical evidence, it sets the stage for transformative, evidence-based planning in Pakistan.

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